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ROLL NO:- 2520030188

SEC :-03

DATABASE:-

```
estore      8      -- DATABASE
            9      DROP DATABASE IF EXISTS bank_db;
lown        10 •    CREATE DATABASE bank_db;
            11 •    USE bank_db;
```

	#	Time	Action
✓	1	20:20:38	DROP DATABASE IF EXISTS bank_db
✓	2	20:20:38	CREATE DATABASE bank_db
✓	3	20:20:38	USE bank_db

1. DDL - DATA DEFINITION LANGUAGE

```
-- 1. CREATE TABLE
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```

```
24 •    DESCRIBE bank_transactions;
25
```

✓	4	20:20:38	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_n
✓	5	20:20:38	DESCRIBE bank_transactions

d System Variables	25	
ort	26	-- 2. ALTER TABLE - ADD COLUMN
ort/Restore	27 •	ALTER TABLE bank_transactions
	28	ADD account_no VARCHAR(20);
Shutdown	29	
gs	30 •	DESCRIBE bank_transactions;
file	31	

✓	6	20:20:38	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)
✓	7	20:20:38	DESCRIBE bank_transactions

les	32	-- 3. ALTER TABLE - MODIFY COLUMN
	33 •	ALTER TABLE bank_transactions
	34	MODIFY customer_name VARCHAR(100);
	35	
	36 •	DESCRIBE bank_transactions;
	37	

✓	8	20:20:38	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)
✓	9	20:20:39	DESCRIBE bank_transactions
✓	10	20:20:39	ALTER TABLE bank_transactions RENAME TO customer_transactions

15

[lections](#)

[Privileges](#)

[System Variables](#)

[:](#)

[t/Restore](#)

37

38 -- 4. ALTER TABLE - RENAME TABLE

39 • ALTER TABLE bank_transactions

40 RENAME TO customer_transactions;

41

42 • DESCRIBE customer_transactions;

--



11 20:29:44 DESCRIBE customer_transactions



12 20:29:45 INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transacti

43

44 -- 5. INSERT VALUE

45 • INSERT INTO customer_transactions

46 (txn_id, customer_name, branch_name, transaction_type, amou

47 VALUES

48 (1, 'Praneeth', 'HITECH CITY', 'CASH', 500.00, '2026-08-10'

49



11 20:32:15 DESCRIBE customer_transactions



12 20:32:15 INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transa

DML - DATA MANIPULATION LANGUAGE

```
-- CREATE TABLE AGAIN FOR DML OPERATIONS
```

```
CREATE TABLE bank_transactions (  
    txn_id INT PRIMARY KEY,  
    customer_name VARCHAR(100),  
    branch_name VARCHAR(50),  
    transaction_type VARCHAR(20),  
    amount DECIMAL(10,2),  
    transaction_date DATE,  
    account_no VARCHAR(20)  
);
```



```
13 20:35:33 SELECT * FROM customer_transactions LIMIT 0, 1000
```

```
INSERT INTO bank_transactions
```

```
(txn_id, customer_name, branch_name, transaction_type, amount, transaction_date,  
VALUES
```

```
(101, 'Praneeth', 'Hyderabad', 'Deposit', 5000.00, '2024-01-10', 'ACC101'),  
(102, 'Sita', 'Hyderabad', 'Withdrawal', 2000.00, '2024-01-11', 'ACC102'),  
(103, 'ANU', 'Vijayawada', 'Deposit', 12000.00, '2024-01-12', 'ACC103'),  
(104, 'SARAT', 'Vizag', 'Deposit', 8000.00, '2024-01-13', 'ACC104'),  
(105, 'VIVAN', 'Hyderabad', 'Withdrawal', 3500.00, '2024-01-14', 'ACC105'),  
(106, 'RAJ', 'Vizag', 'Deposit', 15000.00, '2024-01-15', 'ACC106'),  
(107, 'CHARAN', 'Vijayawada', 'Withdrawal', 1000.00, '2024-01-13', 'ACC107'),  
(108, 'RAM', 'Hyderabad', 'Deposit', 9000.00, '2024-01-14', 'ACC108'),  
(109, 'SEETHARAM', 'Vizag', 'Withdrawal', 4000.00, '2024-01-15', 'ACC109'),  
(110, 'Manikanta', 'Vijayawada', 'Deposit', 11000.00, '2024-01-16', 'ACC110')
```



```
14 20:35:33 TRUNCATE TABLE customer_transactions
```

```
-- 2. UPDATE
UPDATE bank_transactions
SET amount = 5000.00
WHERE txn_id = 105;
```

✓	22	20:35:33	UPDATE bank_transactions SET amount = 5000.00 WHERE txn_id = 105
---	----	----------	--

```
-- 3. DELETE
DELETE FROM bank_transactions
WHERE txn_id = 111;

SELECT * FROM bank_transactions;
```

✓	24	20:35:33	DELETE FROM bank_transactions WHERE txn_id = 111
---	----	----------	--

```

---
113      -- 4. SELECT WITH WHERE CLAUSE
114  ●    SELECT *
115      FROM bank_transactions
116      WHERE transaction_type = 'Deposit';
117
118      -- 5. SELECT WITH ORDER BY
119  ●    SELECT *
120      FROM bank_transactions
121      ORDER BY amount DESC;
122

```

✓	25	20:35:33	SELECT * FROM bank_transactions LIMIT 0, 1000
✓	26	20:35:33	SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 10
✓	27	20:35:33	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000

DCL - DATA CONTROL LANGUAGE



Limit to 1000 rows

- `CREATE USER IF NOT EXISTS 'Auditor1'@'localhost'`
`IDENTIFIED BY 'password123';`
- `GRANT SELECT`
`ON bank_db.bank_transactions`
`TO 'Auditor1'@'localhost';`
- `SHOW GRANTS FOR 'Auditor1'@'localhost';`
- `-- 2. GRANT ALL`
- `CREATE USER IF NOT EXISTS 'BranchManager'@'localhost'`
`IDENTIFIED BY 'password123';`
- `GRANT ALL PRIVILEGES`
`ON bank_db.bank_transactions`
`TO 'BranchManager'@'localhost';`
- `SHOW GRANTS FOR 'BranchManager'@'localhost';`
- `-- 3. REVOKE SELECT`
- `REVOKE SELECT`
`ON bank_db.bank_transactions`
`FROM 'Auditor1'@'localhost';`

⚠	28	20:35:33	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'password123'
✅	29	20:35:34	GRANT SELECT ON bank_db.bank_transactions TO 'Auditor1'@'localhost'
✅	30	20:35:34	SHOW GRANTS FOR 'Auditor1'@'localhost'

TCL - TRANSACTION CONTROL LANGUAGE

- `UPDATE bank_transactions`
`SET amount = 6000.00`
`WHERE txn_id = 101;`
- `COMMIT;`
- `SELECT * FROM bank_transactions;`
- `-- 2. SAVEPOINT`
- `START TRANSACTION;`
- `SAVEPOINT Before_Update;`
- `UPDATE bank_transactions`
`SET amount = 9999.00`
`WHERE txn_id = 102;`
- `ROLLBACK TO SAVEPOINT Before_Update;`
- `COMMIT;`

171

172 • `SELECT * FROM bank_transactions;`

173

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



etup

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	101	Praneeth	Hyderabad	Deposit	6000.00	2024-01-10
	102	Sita	Hyderabad	Withdrawal	9000.00	2024-01-11
	103	ANU	Vijayawada	Deposit	12000.00	2024-01-12
	104	SARAT	Vizag	Deposit	8000.00	2024-01-13
	105	VIVAN	Hyderabad	Withdrawal	5000.00	2024-01-14
	106	RAJ	Vizag	Deposit	15000.00	2024-01-15
	107	CHARAN	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	RAM	Hyderabad	Deposit	9000.00	2024-01-14
	109	SEETHARAM	Vizag	Withdrawal	4000.00	2024-01-15
	110	Manikanta	Vijayawada	Deposit	11000.00	2024-01-16

bank_transactions 107 x

Output



Action Output



	#	Time	Action
✓	1	20:47:45	UPDATE bank_transactions SET amount = 6000.00 WHERE txn_id = 101
✓	2	20:47:48	UPDATE bank_transactions SET amount = 6000.00 WHERE txn_id = 101
✓	3	20:47:50	UPDATE bank_transactions SET amount = 6000.00 WHERE txn_id = 101
✓	4	20:47:53	COMMIT
✓	5	20:47:55	SELECT * FROM bank_transactions LIMIT 0, 1000

```

39      -- 3. FULL TCL EXAMPLE
30 •    START TRANSACTION;
31
32 •    UPDATE bank_transactions
33      SET amount = 7000.00
34      WHERE txn_id = 101;
35
36 •    SAVEPOINT SP1;

```

✓	6	20:49:01	START TRANSACTION
✓	7	20:49:03	UPDATE bank_transactions SET amount = 7000.00 WHERE txn_id = 101
✓	8	20:49:05	UPDATE bank_transactions SET amount = 7000.00 WHERE txn_id = 101
✓	9	20:49:16	UPDATE bank_transactions SET amount = 7000.00 WHERE txn_id = 101
✓	10	20:49:21	SAVEPOINT SP1

Reports

Schema Setup

chemas

```

194      WHERE txn_id = 101;
195
196 •    SAVEPOINT SP1;
197
198 •    UPDATE bank_transactions
199      SET amount = 9000.00
200      WHERE txn_id = 102;
201
202
203 •    COMMIT;
204
205 •    SELECT * FROM bank_transactions;

```

204

205 • `SELECT * FROM bank_transactions;`

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	txn_id	customer_name	branch_name	transaction_type	amount	transaction
▶	101	Praneeth	Hyderabad	Deposit	7000.00	2024-01-10
	102	Sita	Hyderabad	Withdrawal	9000.00	2024-01-11
	103	ANU	Vijayawada	Deposit	12000.00	2024-01-12
	104	SARAT	Vizag	Deposit	8000.00	2024-01-13
	105	VIVAN	Hyderabad	Withdrawal	5000.00	2024-01-14
	106	RAJ	Vizag	Deposit	15000.00	2024-01-15
	107	CHARAN	Vijayawada	Withdrawal	1000.00	2024-01-13
	108	RAM	Hyderabad	Deposit	9000.00	2024-01-14
	109	SEETHARAM	Vizag	Withdrawal	4000.00	2024-01-15
	110	Manikanta	Vijayawada	Deposit	11000.00	2024-01-16

bank_transactions 108 ×

Output



Action Output



	#	Time	Action
✓	10	20:49:21	SAVEPOINT SP1
✓	11	20:50:02	UPDATE bank_transactions SET amount = 9000.00 WHERE txn_id = 102
✓	12	20:50:04	UPDATE bank_transactions SET amount = 9000.00 WHERE txn_id = 102
✓	13	20:50:05	UPDATE bank_transactions SET amount = 9000.00 WHERE txn_id = 102
✓	14	20:50:07	COMMIT
✓	15	20:50:09	SELECT * FROM bank_transactions LIMIT 0, 1000